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The faster or richer the response, the better performance? An empirical analysis of online healthcare platforms from a competitive perspective

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ABSTRACT

The emergence of online healthcare platforms has changed the competitive environment among physicians. However, little is known about how physicians can improve their performance in this new environment. Platforms also face challenges in comprehending the competitive mechanisms among physicians, which might hinder them from formulating strategic managerial decisions that foster sustained growth. In this light, we extract medical service-related information from physicians' response behavioral data on a prominent healthcare platform, and empirically investigate the factors affecting physicians' online performance from a competitive perspective as well as the gender differences in these effects. The results indicate that physicians' responses significantly impact their online performance, revealing a competitive relationship between physicians and their colleagues in the same department. Specifically, a fast response time and informative responses are positively correlated with the focal physician's performance, whereas colleagues' informative responses negatively impact the focal physician's performance, and this relationship is mediated by the focal physician's response informativeness. Nevertheless, there is no significant correlation between colleagues' response time and the focal physician's performance. The results also unveil that gender moderates the effect of response informativeness on the focal physician's performance. Specifically, colleagues' response informativeness has a more significant impact on male physicians' performance than on female physicians' performance, suggesting a greater propensity for competition among male physicians. Our findings could offer decision support for enhancing physician performance and healthcare platform management.

1. Introduction

The successful development of online healthcare platforms (OHPs) is transforming how physicians deliver services to patients from an offline format to a combination of online and offline formats [1]. Notably, OHPs have overcome the temporal and geographical limitations of traditional medical services [2,3]. They serve as a convenient channel for physician–patient communication [4,5], allowing for more flexible scheduling of medical consultations and enabling physicians to provide services nationwide at any time. This change, however, has also introduced new patterns of competition among physicians. On the one hand, the boundaries of competition have broadened as OHPs expose physicians to a larger pool of competitors nationwide. Specifically, offline competition often occurs among physicians in the same hospital or city, while OHPs expose physicians to competition with other physicians across the country. On the other hand, the structure, relationships, and mechanisms characterizing competition among physicians have also gradually undergone transformation. In light of such changes in physician competition, it is worth exploring how physicians can make decisions to enhance their performance in this new competitive environment.

A group of studies on how physicians achieve good performance has produced several insights. Researchers have found that physicians' social and economic returns in OHPs are positively associated with the interaction between their status and decisional capital [6]. Additionally, a physician's online and offline reputation positively affects his or her appointment load [7]. Despite these insights, the dominant view of factors influencing online performance has been primarily physician-

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centric, emphasizing individual behaviors or characteristics [8,9], such as reputation, service quality, title, experience, and breadth and depth of services. The existence of competition among colleagues providing similar medical services has not been fully considered. Given that many physicians can deliver medical services, a comprehensive understanding of online performance necessitates attention to competition, which can both constrain and motivate the response efforts of physicians. Although previous studies on performance have explored competition in e-commerce [10–12], importantly, online healthcare services differ from e-commerce services. Specifically, due to the diverse nature of patients' conditions and problems, online healthcare services, in which physicians can attain a competitive advantage through their efforts, emphasize on providing personalized care to each patient. In contrast, the competitive strategies of e-commerce companies are very different. Related research has been oriented more toward online searches, where competition is concentrated on company brand reputation [13], price [14], product diversity, product description, and web design [15], among others. Hence, based on these differences, the findings of previous studies already conducted on the topic of e-commerce may no longer apply in the context of online medical services.

A few scholars have studied the nature of competition in healthcare markets and its effect on quality [16,17] and social welfare [18] using data on offline care from nursing homes, insurers, and hospitals. Some scholars have also demonstrated that there is competition among physicians [19]. Various studies have investigated price competition in e-commerce [20,21]. However, due to the emergence of online competition among physicians and the significant differences between e-commerce and online medical services, the findings of existing studies cannot explain how physicians, including a particular physician (referred to as the "focal physician") and all other physicians in the same specialty on the platform (referred to as "peer physicians" or "colleagues"), compete in the online environment and the mechanisms by which they compete. In other words, in the new online competitive environment, how physicians can gain a competitive advantage and thus improve their online performance remains unclear.

The literature on satisfaction in healthcare reveals that physicians' responses significantly impact patients' perceptions and adoption of healthcare services [22] and, consequently, recommendations for physicians [23]. A physician's response can serve as a signal of quality and reputation and play a crucial role in conveying service-related information to patients who seek professional assistance on online health platforms. Hence, investigating the features of physicians' online responses may benefit the responding physicians and improve their performance. Accordingly, we seek to extract service-related information from physicians' response behavior and determine the impact of physicians' responses on their online performance. Typically, physicians respond promptly and thoughtfully online, not only to engage in social interaction with patients but also to signal their service quality and give patients additional information. In our context, a good physician can effectively signal the quality of healthcare services to patients through timely and thoughtful online responses, potentially earning greater recognition from patients and achieving better performance.

Considering online responses as a crucial determinant of online performance [24], our work investigates whether physician response efforts, encompassing response time and response informativeness, affect online performance from a competitive perspective. Specifically, our study conceptualizes response efforts as a behavioral contest, focusing on the impact of differences in response time and informativeness between the focal physician and peer physicians, and explores mechanisms of competition among physicians in the online environment.

In addition, social research has placed great emphasis on gender differences. Much of the literature regarding the impact of gender differences on performance has highlighted organizational performance [25] and individual performance [26]. According to a study by Roter et al. [27] on physician–patient communication, female physicians spend an average of 10% more time than their male counterparts listening to patients, ultimately making them more profitable. In a study on gender differences in physician communication styles, Barr [28] observed that male physicians are often notorious for frequently interrupting patients. Moreover, on average, medical visits with female physicians are 2 min longer than those with male physicians. Consequently, patients are more inclined to trust female physicians and seek treatment from them, resulting in higher performance and a more competitive advantage for female physicians. Meanwhile, previous findings in experimental and personnel economics have analyzed gender differences along with competition and come to a common conclusion that men prefer competition [29,30]. Regarding personality characteristics, women possess higher levels of collaborative personality traits than men [31]. These findings from various studies regarding gender differences inspire us to explore the moderating effect of gender on physicians' online healthcare services, given that the moderating role of gender in the relationship between service behavior and performance remains unclear. Thus, the research questions addressed in this study are as follows:

(1) How do the focal physician's response efforts affect his or her online performance?
(2) How do peer physicians' response efforts affect the focal physician's online performance?
(3) How does gender moderate the relationship between physicians' response efforts and their online performance?

In this research, all consultation data were collected from a leading physician–patient healthcare platform in China in 2017. The dataset included data from 34,269 physicians and covered 3,469,809 consultation orders. Data cleaning and collation were carried out using Python, and the final panel data were obtained by counting and calculating variables for physicians over natural weeks. The empirical findings reveal that physicians' responses influence their online performance and the competitive relationship between the focal and peer physicians. The results show that the faster the response time or the richer the response informativeness of the focal physician is, the better the focal physician's performance, whereas the richer the peer physicians' responses are, the worse the focal physician's performance. Nonetheless, peer physicians' response time has no significant effect on the online performance of the focal physician, a finding that can be partly attributed to the online communication environment, patients' health conditions, and their expectations. Additionally, the response informativeness of the focal physician mediates the relationship between the response informativeness of peer physicians and the focal physician's performance. More importantly, our results suggest that gender plays a moderating role between the response informativeness of peer physicians and the focal physician's online performance.

This paper presents three theoretical contributions as follows. First, our research enriches the research stream on online performance by revealing the effect of physician response efforts on online performance from a competitive perspective. The analysis extends beyond individual factors, contributing to a more comprehensive understanding of the determinants of physicians' online performance. We find that the more effort physicians put into their responses, the better their performance is. The response efforts of both the focal physician and his or her colleagues are factors that influence the focal physician's online performance. Second, this study expands and complements the research on competition theory among physicians in the online healthcare community by considering the mediating effects of the focal physician's response efforts between colleagues' response efforts and their performance, and concludes that a competitive relationship exists among physicians on the platform. Additionally, our work extends relevant prior work on competition in e-commerce and offline contexts to the context of online healthcare services. Third, our work advances the research on gender differences in online services by examining how gender moderates the association between physicians' response efforts and online performance. Consistent with prior studies [29,30], our study provides evidence that male physicians prefer competition. An additional essential finding is that the influence of colleagues' response informativeness on the performance of male focal physicians is greater than that of female focal physicians. That is, male physicians have a less competitive advantage than females.

Our work also has important practical contributions. First, to achieve better performance, physicians should make decisions that enhance response speed or increase the informativeness of their responses. They should also monitor the response informativeness of their peers on the platform to compete effectively and make their own response decisions based on their colleagues' responses. Our findings may provide useful guidelines to help physicians decide how to improve their performance. Second, platforms should decide to add a "response efforts index" on each physician's homepage, which guides patients to make better decisions about the adoption of physicians and helps physicians adjust and improve their response behavior and adjust their decision-making according to their actual situation. Third, the heterogeneity test results for moderating effects suggest that physicians should decide to upload their personal pictures if they prefer to compete. Fourth, although our findings are based on the healthcare context, they can be applied to other similar service industries, such as open-source communities.

2. Literature review

Our study is related to three research streams: (1) online healthcare services, (2) competitive effects in online services, and (3) gender effects in online services.

2.1. Online healthcare services

As a supplement to offline services [1], online services can help physicians track patients' conditions and increase their own performance [6,32]. Some scholars have explored the impact of factors such as physicians' title [33], service quality [8], online and offline reputation [7], and experience [23] on online performance. Other scholars have proposed that disease knowledge [7], privacy concerns [34], waiting time [24,35], household net income category, and education level [36] all influence patients' decision making, thereby affecting physicians' online performance. The literature has demonstrated from diverse perspectives that various factors in online medical services affect physicians' online performance. However, limited literature has focused on physicians' response efforts from a competitive perspective. In this study, we will focus on two dimensions, namely, physicians' response time and response informativeness, to investigate their causal effects on physicians' online performance.

2.2. Competitive effects in online services

The enormous potential benefits of the online service market make the competition among sellers increasingly fierce. Thus, there is an urgent need to understand the mechanisms of competition among sellers to help them compete with their rivals and survive in the market environment. For example, Li et al. [37] explain sellers' performance in e-marketplace competition to understand the impact of online service quality and use competitive dynamics theory to conclude that sellers who offer many more complex and diverse platform-based services perform better in the e-marketplace. Chang et al. [38] study how online travel agency (OTA) websites and hotel websites compete to attract customers in a multichannel environment. Wu and Lu [39] study how the reputation of the focal physician and his or her colleagues influences the number of reviews he or she receives in the future. Rossi [10] investigates how competition affects the role of reputation in incentivizing sellers to put in effort and finds that homeowners' efforts decrease when the number of competitors increases.

Competition represents a process whereby healthcare service providers strive to achieve better outcomes. Therefore, it is essential to comprehend the mechanisms of competition in the operation of the healthcare services market. However, previous literature on the online competitive relationship among physicians is limited and primarily centers on price and service quality. Studies concerning the terms "focal" and "peer" have predominantly focused on peer effects or social effects, rather than competitive effects. This study explores the competitive relationship between the focal physician and colleagues in the same specialty regarding online response efforts.

2.3. Gender effects in online services

Some studies have focused on gender differences in user behavior on online platforms [40]. For example, Vasilescu et al. [41] evaluate the presence of females in a highly skilled software-development-related online Q&A platform and compare the frequency and duration of their participation with that of their male counterparts. Moreover, Kimbrough et al. [42] find that compared to men, women typically use online technology more frequently and show a greater preference for indirect communication. Wang et al. [43] find that men are community members for longer periods in an online travel community.

Some research has also been conducted on gender differences in competition preferences [30,44]. For example, Gneezy et al. [45] find that men's performance is higher than women's in competitive environments. Günther et al. [46] find that the level of improvement of male labor output in competitive settings is significantly higher than that of female labor output. Meanwhile, some studies have directly compared competition preferences between men and women to explain gender differences. For example, Niederle and Vesterlund [47] find that the number of women willing to compete is significantly lower than the number of men and that there are significant gender differences in competition preferences. Beckmann and Menkhoff [48] find that female managers are more averse to competition than male managers.

However, there is less empirical research on online competition among healthcare professionals and potential gender differences. Based on the literature, we concentrate on the impact of gender on the correlation between physician response efforts and online performance on healthcare platforms.

3. Hypotheses

Online reputation, influenced by the responsiveness and interactions of users [49], becomes an essential basis for consumers to make purchase decisions [50]. The faster or the richer online Q&A responses are, the better an online reputation can be established. Strangers feel more connected when their conversation partners respond quickly [51]. In online healthcare consultations, physicians build their online reputation by answering quickly and providing a wealth of information to promptly and effectively address patients' online queries [33], which positively impacts their performance.

On a healthcare platform, patients can observe their physician's online efforts through online consultation details, such as informativeness and response time. Physicians can influence consumer perceptions by actively responding to patients and interacting with them. Patients can intuitively perceive physicians' commitments and effort by viewing their responses and published articles. These perceptions subsequently shape their trust in the physician, which affects their decision making. Therefore, richer response informativeness and faster response times, which indicate greater online efforts, lead to the development of a better online reputation, which in turn has a positive impact on the physician's acquisition of new consultations and service revenue. Thus, we propose the following:

Hypothesis 1A. (H1A): The focal physician's response time negatively and significantly affects online performance.

Hypothesis 1B. (H1B): The focal physician's response informativeness positively and significantly affects online performance.

On OHPs, physicians compete fiercely in the same category of services [52,53]. On the one hand, the number of physicians providing online services is increasing. Platforms encourage physicians to make themselves more competitive in the market [54]. On the other hand, the boundaries of competition among physicians have been broken down, and the competitive relationship has changed. Instead of competing with colleagues in the same hospital or city, as was traditionally the case in offline care, each physician has to compete with physicians in the same specialty across the country on OHPs. The faster or richer the responses from peer physicians are, the greater the reputation they build and the stronger their competitive advantage, which decreases the focal physician's online performance. Consequently, we may infer the following:

Hypothesis 2A. (H2A): Peer physicians' average response time positively and significantly affects the focal physician's online performance.

Hypothesis 2B. (H2B): Peer physicians' average response informativeness negatively and significantly affects the focal physician's online performance.

We suggest that the focal physician's response efforts mediate the effects of peer physicians' response efforts on his or her performance. To survive in a highly competitive environment, an active approach is needed [55]. In the context of e-healthcare, response quality is important for patients. Physicians engage in competition to respond to patient inquiries quickly and comprehensively. A faster response time and richer response informativeness may attract a larger patient base. When the focal physician observes his or her peers responding more swiftly and with more comprehensive information than he or she does, he or she is motivated to elevate his or her response speed and offer more detailed information to attract more patients and thereby achieve higher performance. Consequently, we conclude the following:

Hypothesis 3A. (H3A): The focal physician's response time mediates the relationship between peer physicians' average response time and his or her online performance.

Hypothesis 3B. (H3B): The focal physician's response informativeness mediates the relationship between peer physicians' average response informativeness and his or her online performance.

The literature has argued that men are more likely to overestimate their achievements than women and place too much weight on their probability of winning, which leads to too many men choosing to compete and more women giving up in competition [56]. Additionally, risk decision theory holds that women are more risk averse than men and that such differences naturally promote gender differences in competition preferences [47]. In addition to by sociocultural factors, innate characteristics such as genes influence gender differences in competition preferences. Genetic or hormonal differences can make women less competitive than men [57]. Thus, we argue that gender also moderates the relationship between focal physicians' responses and their online performance in online health services.

Furthermore, compared to men, women adopt a more patient-centered communication style with longer consultation sessions [27], adhere more closely to clinical protocols, provide preventive care more frequently [28], and offer more psychosocial counseling. In particular, women also tend to disclose more personal information during conversations, exhibit a warmer and more engaged nonverbal communication style, and encourage and enable others to engage in free and intimate communication with them [27]. Consequently, due to these inherent characteristics, the response time or response informativeness of female focal physicians may be less affected. Thus, the performance of the female focal physician is less susceptible to peer physicians' responses. Therefore, we infer that peer physicians' responses affect male focal physicians' performance more than female focal physicians' performance. Thus, we posit the following hypothesis:

Hypothesis 4A. (H4A): Gender moderates the relationship between physician response time and online performance such that, compared with women, men have better performance when responding faster.

Hypothesis 4B. (H4B): Gender moderates the relationship between physician response informativeness and online performance such that, compared with women, men have better performance when responding richer.

In summary, Fig. 1 illustrates our research framework.

4. Data and methodology

4.1. Research context

We collected data from a leading OHP in China founded in 2006. As of 2022, over 260,000 physicians had registered on this platform to provide online medical services to more than79 million patients, with 73% of them from Class A tertiary hospitals. Physicians are required to provide their medical practitioner certificate, as well as background information about their workplace, gender, title, and specialty area when registering, and only when verified by the platform can they provide medical services to patients. Through this platform, physicians can provide several services, such as online consultation, appointment referrals, information inquiry, and disease knowledge sharing. Importantly, patients pay physicians directly for paid consultations. Fig. 2 provides a screenshot of the main functionalities of the platform.

We gathered physicians' individual information, such as physician ID, specialty, title, number of articles published, reviews, virtual gifts, and detailed physician–patient consultation record data by Python. They were linked together using the physician ID. We manually categorized the physicians into eight departments by interviewing them (see Fig. 3) and selected three specialist physicians for review and validation. For physician–patient consultation (one-on-one) record data, we performed an analysis of text structure and semantic features to separate each consultation into physicians' responses and patients' inquiries. Subsequently, based on the response dates, we extracted responses corresponding to the initial day of inquiry from the physician's replies. Finally, we removed irrelevant text and stop words by Jieba, considering the processed text content as valid responses from the physician.

Due to the limited timeframe for patients to receive medical attention, we excluded consultations where the physician's first response exceeded 30 days from when the patient posed the question. After data processing, there were 3,469,809 consultations from 34,269 physicians, of whom 64.01% were male and 35.99% were female. The observation period lasted from January 2, 2017, to December 31, 2017, spanning 52 weeks. All variables were standardized by z-score for analysis.

4.2. Variable measurement

The dependent variable in our research is *physician performance* ($VolPay_{it}$), often measured by the total number of paid consultations the physician responded to [58–60]. In addition, considering that the number of paid consultations by a physician in a short period may be too small, or even have a large number of 0. It may not be able to accurately portray the physician's performance, which is not conducive to estimating model parameters. Therefore, this study followed the natural week for the data statistics.

The independent variables in this study include *response time* and *response informativeness* from both the focal physician and peer physicians. We measured *response time* by calculating the average response time between the patient's first question and the physician's first response [61]. *Response informativeness* refers to the extent to which service providers offer detailed information, such as diagnostic results

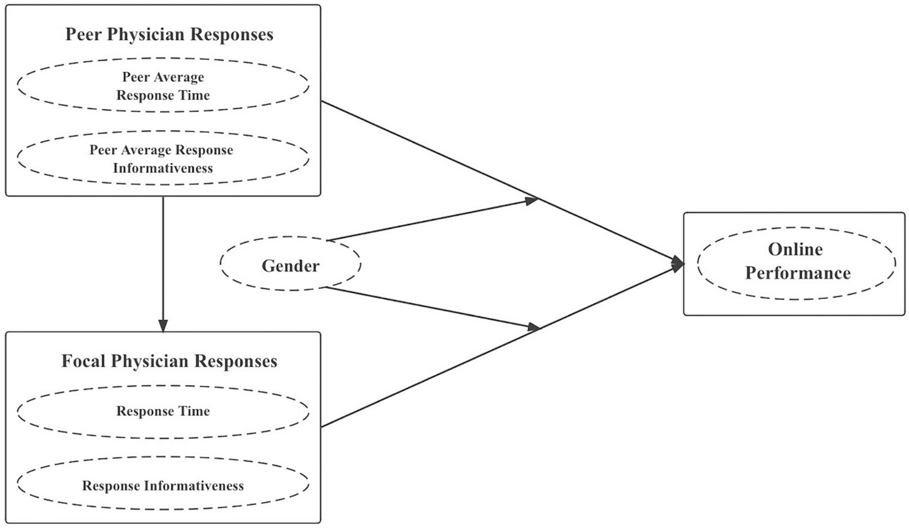

Fig. 1. Research framework.

Fig. 2. Screenshot of the main functionalities on the platform.

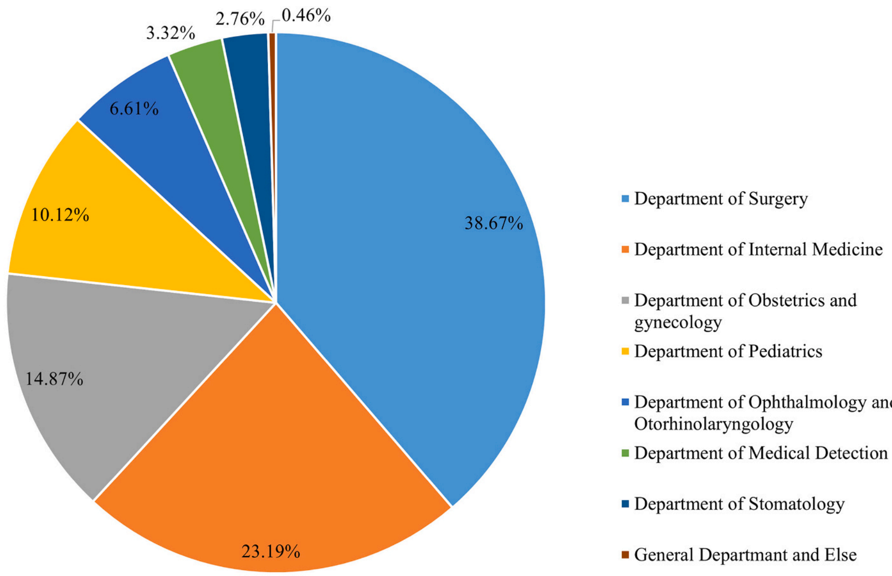

Fig. 3. Distribution of physicians in the departments.

[62]. In online consultations, physicians deliver health information to patients through textual responses [63]. And text length reflects the depth and comprehensiveness [64], serving as a manifestation of quality. According to Chen et al. [63], we utilize the text from online physician-patient consultation records to assess the *response informativeness* using natural language processing techniques and measured *response informativeness* by determining the average number of words for each consultation after removing of stop words. The four independent variables are calculated as follows:

$$FocResTime_{it} = \frac{\sum_{i\in\mathcal{I},k\in\mathcal{K},t\in\mathcal{T}} x_{ikt}}{N} = \frac{\sum_{i\in\mathcal{I},k\in\mathcal{K},t\in\mathcal{T}} answertime_{ikt} - askingtime_{ikt}}{N} \tag{1}$$

$$FocResInfo_{it} = \frac{\sum_{i\in\mathcal{I},k\in\mathcal{K},t\in\mathcal{T}} z_{ikt}}{N} = \frac{\sum_{i\in\mathcal{I},k\in\mathcal{K},t\in\mathcal{T}} answerinfo_{ikt} - askinginfo_{ikt}}{N} \tag{2}$$

$$PeerResTime_{it} = \frac{\sum_{i\neq j\in\mathcal{I},\mathcal{D}_i=\mathcal{D}_j,k\in\mathcal{K},t\in\mathcal{T}} FocResTime_{jt}}{M-1} = \frac{\sum_{i\neq j\in\mathcal{I},\mathcal{D}_i=\mathcal{D}_j,k\in\mathcal{K},t\in\mathcal{T}} \frac{\sum x_{ikt}}{N}}{M-1} \tag{3}$$

$$PeerResInfo_{it} = \frac{\sum_{i\neq j\in\mathcal{I},\mathcal{D}_i=\mathcal{D}_j,k\in\mathcal{K},t\in\mathcal{T}} FocResInfo_{jt}}{M-1} = \frac{\sum_{i\neq j\in\mathcal{I},\mathcal{D}_i=\mathcal{D}_j,k\in\mathcal{K},t\in\mathcal{T}} \frac{\sum z_{ikt}}{N}}{M-1} \tag{4}$$

where x_{ikt} and z_{ikt} represent the response time and response informativeness for physician i's k^{th} consultation in week t. M is the total number of physicians in the same department, and the number of consultations responded to by physician i in week t is N. $\mathcal{I}$, $\mathcal{D}$, $\mathcal{K}$, and $\mathcal{T}$ are the physician set, department set, consultation set, and time set, respectively.

Our moderator was gender. The gender of the physician is not specifically indicated on haodf.com. Some of them are mentioned in their profile on their homepage. We first extract the gender characteristics of physicians by text recognition. For physicians who are not marked with gender, we mark them manually based on the photos provided by physicians. If the physician's gender was male, $Male = 1$; otherwise, $Male = 0$.

We also had several control variables, including the number of online reviews received by physicians (NOR_{it}), the number of virtual gifts from patients (NOG_{it}), the number of online articles published (NOA_{it}), the

Table 1
Descriptive statistics.

Variables	Definition	Mean	SD	Min	Max
VolPay	The total number of online paid consultations each physician each week	1.84	2.25	1	68
PeerResTime	The average response waiting time of peer physicians (in the same department as the focal physician) each week	0.54	0.19	0	3.33
PeerResInfo	The average response informativeness (average number of response words) of peer physicians (in the same department as the focal physician) each week	108.82	48.03	0	384.67
FocResTime	The average response waiting time of the focal physician each week	0.54	1.09	0	29.75
FocResInfo	The average response informativeness of the focal physician each week	108.83	119.75	0	1095
Free	The total number of free online consultations answered by each physician each week	4.15	11.09	0	198
Price	The average price of consultations answered by each physician each week	31.66	40.07	1	1794
NOR	The total number of patients' online reviews received by each physician each week	0.70	1.65	0	89
NOG	The total number of virtual gifts given to the physicians each week	0.51	2.74	0	244
NOA	The total number of online articles posted by each physician each week	0.074	0.74	0	121
Male	The gender of the focal physician (male encode as 1, female encode as 0)	0.64	0.48	0	1

Table 2
Matrix of correlation coefficient.

Variables	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
(1) VolPay	1.000										
(2) FocResTime	−0.045	1.000									
(3) FocResInfo	−0.004	−0.102	1.000								
(4) PeerResTime	0.008	0.162	−0.256	1.000							
(5) PeerResInfo	−0.024	−0.111	0.397	−0.640	1.000						
(6) Free	0.059	−0.108	0.194	−0.184	0.204	1.000					
(7) NOA	0.007	−0.028	0.046	−0.011	0.013	0.064	1.000				
(8) NOG	0.131	−0.042	0.048	−0.024	0.042	0.063	0.039	1.000			
(9) NOR	0.225	−0.083	0.010	0.006	−0.004	0.070	0.036	0.253	1.000		
(10) Price	0.145	−0.044	0.162	−0.048	0.074	−0.015	0.006	0.127	0.230	1.000	
(11) Male	0.026	−0.014	0.016	0.011	0.003	0.055	0.028	0.039	0.073	0.005	1.000

number of free consultations offered ($Free_{it}$), and the price of consultations ($Price_{it}$).

Table 1 presents the main variables and the descriptive statistics. Table 2 provides the correlation coefficient matrix between variables. The largest correlation (0.640) between different variables occurs between *PeerResTime* and *PeerResInfo*, but this is not sufficient to induce multicollinearity issues [63].

4.3. Empirical model

This research aimed to better understand how physician response efforts affect the focal physician's online performance. We first introduce the main effect model, which addresses the impacts of (1) the focal physician's responses and (2) peer physicians' responses on a focal physician i's performance in the following regression model:

$$\begin{aligned} VolPay_{it} = {} & \lambda_0 + \lambda_1 \mathrm{Peer}ResTime_{it} + \lambda_2 \mathrm{Peer}ResInfo_{it} + \lambda_3 FocResTime_{it} \\ & + \lambda_4 FocResInfo_{it} + \lambda_5 Free_{it} + \lambda_6 Price_{it} + \lambda_7 NOR_{it} + \lambda_8 NOG_{it} \\ & + \lambda_9 NOA_{it} + \lambda Week_t + \gamma_i + \varepsilon_{it} \end{aligned} \tag{5}$$

where all the control variables are time-variant. To capture unobserved heterogeneity, we include individual fixed effects (γ_i, *IFE*) and time ($Week_t$, *TFE*) fixed effects. λ_0 to λ_9 are the parameters to be estimated, and ε_{it} is the error term.

In Eq. (6), we examine how gender moderates the impact of physicians' responses on their online performance:

$$\begin{aligned} VolPay_{it} = {} & \lambda_0 + \lambda_1 \mathrm{Peer}ResTime_{it} + \lambda_2 \mathrm{Peer}ResInfo_{it} + \lambda_3 FocResTime_{it} \\ & + \lambda_4 FocResInfo_{it} + \theta_1 \mathrm{Peer}ResTime_{it} \times Male_i + \theta_2 \mathrm{Peer}ResInfo_{it} \\ & \times Male_i + \theta_3 FocResTime_{it} \times Male_i + \theta_4 FocResInfo_{it} \times Male_i \\ & + \lambda_5 Free_{it} + \lambda_6 Price_{it} + \lambda_7 NOR_{it} + \lambda_8 NOG_{it} + \lambda_9 NOA_{it} + \lambda Week_t \\ & + \gamma_i + \varepsilon_{it} \end{aligned} \tag{6}$$

where $Male_i$ is the indicator of physician i's gender. The interaction coefficients, θ_1 to θ_4, capture the effect if any, of physician gender on focal physician i's online performance.

We also test the issue of multicollinearity in regression Eqs. (5) and (6) by calculating each variable's variance inflation factor (VIF) and tolerance, which is the reciprocal of the VIF. We find that the VIFs associated with each variable range from 1.01 to 1.72, less than the general cutoff value of 10 (VIF<10), indicating that multicollinearity is not an issue in our model.

5. Results

5.1. Main effects

In Table 3, Model 2 shows that the coefficient of *FocResInfo* is positive and significant, while the coefficients of *PeerResInfo* and *FocResTime* are both negative and significant. These results support H1A, H1B, and H2B, indicating that the focal physician's response efforts and peers' informativeness are essential factors in determining the focal physician's performance. Thus, the faster and more informative the physician's responses are, or the poorer colleagues' responses are, the more online paid consultations the focal physician receives.

Table 3
Results for Main Effects.

Variables	Model 1 VolPay	Model 2 VolPay
Free	−0.0115	−0.0110
	(−0.76)	(−0.72)
NOA	0.00545***	0.00523**
	(2.59)	(2.52)
NOG	0.00654	0.00637
	(0.65)	(0.63)
NOR	0.0329***	0.0326***
	(4.46)	(4.42)
Price	−0.0637***	−0.0678***
	(−7.56)	(−7.45)
FocResTime		−0.0139***
		(−7.92)
FocResInfo		0.00954***
		(3.07)
PeerResTime		−0.00273
		(−0.19)
PeerResInfo		−0.152***
		(−5.13)
TFE	YES	YES
IFE	YES	YES
_cons	−0.00686**	0.142***
	(−0.66)	(4.17)
Observations	196,570	196,570

Notes: t statistics in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Notably, the coefficient of *PeerResTime* is insignificant, rejecting H2A. Thus, the average response time of peer physicians has no significant effect on online performance. A possible reason is that even though rapid responses help physicians establish a better reputation and become more competitive in the market, patients seeking online consultations are typically not experiencing acute medical conditions, and they have the patience to wait for more professional, detailed, and accurate responses or even more in-depth explanations from their physicians. Simultaneously, patients may have varied expectations when using online consultation services. However, their foremost expectation is that physicians can provide disease diagnoses and treatment recommendations. Patients understand that online platforms typically differ from offline visits, where physicians can respond immediately. Therefore, while response time does impact the perception of online service quality, patients are still willing to accept longer response times in exchange for higher-quality and more meticulous answers. In other words, patients have a higher tolerance for waiting time. Therefore, in an online medical platform that allows patients to choose their physicians, peer

Table 4
Results for the Mediating Effects of Focal Physician's Responses.

Variables	Model 1	Model 2	Model 3	Model 4	Model 5	Model 6	Model 7
	FocResTime	FocResInfo	VolPay	VolPay	VolPay	VolPay	VolPay
Free	−0.000445	0.0804***	−0.0102	−0.0122	−0.0102	−0.0110	−0.0110
	(−0.16)	(14.84)	(−0.67)	(−0.80)	(−0.67)	(−0.72)	(−0.72)
NOA	−0.00561***	0.0113***	0.00541***	0.00526**	0.00533**	0.00530**	0.00523**
	(−3.05)	(4.32)	(2.58)	(2.53)	(2.56)	(2.54)	(2.52)
NOG	−0.00647**	0.00702**	0.00653	0.00639	0.00644	0.00646	0.00637
	(−2.04)	(2.10)	(0.65)	(0.63)	(0.64)	(0.64)	(0.63)
NOR	−0.0183***	−0.00308	0.0328***	0.0327***	0.0325***	0.0328***	0.0326***
	(−7.78)	(−1.14)	(4.45)	(4.43)	(4.41)	(4.46)	(4.42)
Price	−0.00223	0.430***	−0.0637***	−0.0678***	−0.0637***	−0.0678***	−0.0678***
	(−0.62)	(21.16)	(−7.57)	(−7.43)	(−7.57)	(−7.45)	(−7.45)
PeerResTime	−0.00363		−0.00261		−0.00265	−0.00269	−0.00273
	(−0.35)		(−0.18)		(−0.18)	(−0.19)	(−0.19)
PeerResInfo		0.0601***	−0.152***		−0.152***	−0.153***	−0.152***
		(3.08)	(−5.12)		(−5.11)	(−5.14)	(−5.13)
FocResTime				−0.0140***	−0.0140***		−0.0139***
				(−7.96)	(−7.96)		(−7.92)
FocResInfo				0.00926***		0.00968***	0.00954***
				(2.98)		(3.12)	(3.07)
TFE	YES	YES	YES	YES	YES	YES	YES
IFE	YES	YES	YES	YES	YES	YES	YES
_cons	−0.141***	0.301***	0.148***	−0.0122***	0.145***	0.145***	0.142***
	(−10.07)	(12.19)	(4.32)	(−1.16)	(4.25)	(4.24)	(4.17)
Observations	196,570	196,570	196,570	196,570	196,570	196,570	196,570

Notes: t statistics in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

physicians' short average response time does not threaten the number of online paid consultations for the focal physician.

5.2. The mediating roles of the focal physician's responses

We conduct mediation analyses following the procedures outlined by Baron and Kenny [65], and the results are shown in Table 4. Models 1 and 2 show that *PeerResTime* has no significant effect on *FocResTime*, while *PeerResInfo* significantly positively impacts *FocResInfo*. The reason why *PeerResTime* is insignificant may be that physicians are overloaded with offline work. Meanwhile, physicians' time and energy are limited, so the speed of their online responses is not something they can control. As a result, when peer physicians reply quickly, even if the focal physician wishes to do so, he or she is often "overwhelmed" and unable to reply in time.

Model 3 shows that *PeerResInfo* significantly affects online performance, while the effect of *PeerResTime* is insignificant. The results from Model 4 show that *FocResTime* has a negative effect on the focal physician's performance and that *FocResInfo* has a positive impact. To facilitate testing of the different mediating effects of the focal physician's response time and informativeness, we regress online performance on *FocResTime* and peer physicians' responses in Model 5, which shows that *FocResTime* and *PeerResInfo* are significant, while *PeerResTime* is not. Similarly, we regress online performance on *FocResInfo* and peer physicians' responses in Model 6, showing that *FocResInfo* and *PeerResInfo* are significant, while *PeerResTime* is not. Finally, we regress online performance on the focal and peer physicians' responses. As shown in model 7, all variables except *PeerResTime* are significant, and these results suggest that *FocResInfo* plays a partially mediating role. The implication is that the informativeness of the responses of the focal physician acts as a mediator between the informativeness of colleagues' responses and online performance, and the effect is transmitted through competition between the focal and peer physicians. In summary, the results reject H3A and support H3B.

5.3. The moderating role of gender

In this section, we examine how gender moderates the impact of physicians' responses. Following Toothaker [66], we center the variables to reduce possible problems of multicollinearity. The results are shown in Table 5.

Table 5
Results for the moderating effects of gender.

Variables	Model 1	Model 2	Model 3	Model 4
	VolPay	VolPay	VolPay	FocResInfo
Free	−0.0115	−0.0110	−0.0101	0.0800***
	(−0.76)	(−0.72)	(−0.67)	(14.76)
Price	−0.0637***	−0.0678***	−0.0679***	0.430***
	(−7.56)	(−7.45)	(−7.47)	(21.16)
NOR	0.0329***	0.0326***	0.0325***	−0.00303
	(4.46)	(4.42)	(4.41)	(−1.13)
NOG	0.00654	0.00637	0.00656	0.00695**
	(0.65)	(0.63)	(0.65)	(2.08)
NOA	0.00545***	0.00523**	0.00530**	0.0113***
	(2.59)	(2.52)	(2.55)	(4.30)
stdPeerResTime		−0.00273	−0.00124	
		(−0.19)	(−0.09)	
stdPeerResInfo		−0.152***	−0.119***	0.0463***
		(−5.13)	(−4.14)	(2.34)
stdFocResTime		−0.0139***	−0.0168***	
		(−7.92)	(−6.06)	
stdFocResInfo		0.00954***	0.00779*	
		(3.07)	(1.96)	
stdPeerResTime × *Male*			−0.00395	
			(−0.46)	
stdPeerResInfo × *Male*			−0.0456***	0.0188***
			(−5.32)	(3.05)
stdFocResTime × *Male*			0.00513	
			(1.45)	
stdFocResInfo × *Male*			0.00319	
			(0.65)	
TFE	YES	YES	YES	YES
IFE	YES	YES	YES	YES
_cons	−0.0248**	0.124***	0.119***	0.319***
	(−2.38)	(3.65)	(3.49)	(17.56)
Observations	196,570	196,570	196,570	196,570

Notes: t statistics in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

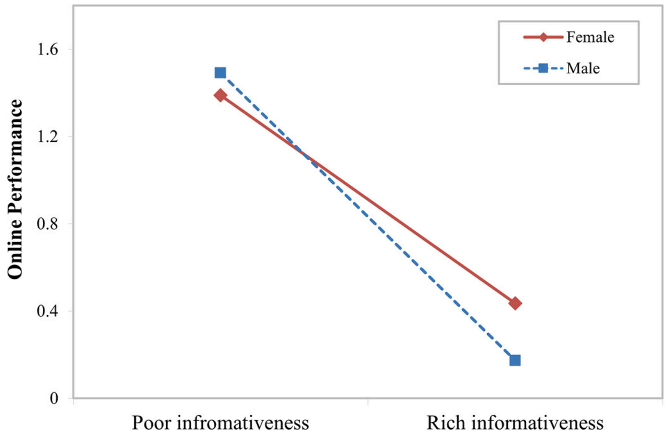

Fig. 4. Interaction effects of gender.

In Model 3, it can be found that only the coefficient of *stdPeerResInfo* × *Male* is significantly negative, indicating that gender reinforces the negative relationship between the informativeness of peer physicians' responses on the focal physician's performance. In other words, for male physicians (*Male* = 1), colleagues' response informativeness has a more significant impact on the focal physician's performance. The results for *stdPeerResTime* × *Male*, *stdFocResTime* × *Male*, and *stdFocResInfo* × *Male* are all insignificant, suggesting that gender does not moderate the effects of the focal physician's response efforts (time and informativeness) and the average response time of peer physicians on performance, which reject H4A. Model 4 shows that male physicians prefer competition in terms of the informativeness of their responses. Nonetheless, Model 3 indicates that male physicians have less of a competitive advantage than female physicians, partially supporting H4B.

Fig. 4 illustrates the moderating effect of gender on the relationship between the informativeness of peer physicians' responses and the focal physician's performance, indicating downward trends for both male and female physicians, suggesting that *PeerResInfo* has a positive incentive effect on online performance. However, the slope of the corresponding line for male physicians is significantly greater than that for female physicians, which means that for male physicians (*Male* = 1), the incentive effect of *PeerResInfo* on performance is more pronounced.

5.4. Robustness checks

To ensure the robustness of these findings, we conducted robustness checks from two perspectives: model substitution and variable substitution. We also performed heterogeneity tests based on different departments and whether physicians uploaded pictures to validate our findings.

5.4.1. Model substitution

First, different models were selected for further testing. We added department-specific linear trends into our model. Given that our dependent variable, the number of the focal physician's online paid consultations, takes on nonnegative integer values, we also ran count data models and a log-transformation model as robustness checks. As shown in Table 6, the results are consistent with the main results.

5.4.2. Variable substitution

In addition, we recalculated *PeerResTime* and *PeerResInfo*. Firstly, considering that regional healthcare quality is related to the level of regional economic development, regions with similar levels of economic development also tend to have comparable levels of healthcare quality. Therefore, we categorized physicians within the same department located in the same region as the focal physician, based on the level of economic development in their respective areas, as competitors to the focal physician. We used two criteria to divide the regions. One was the eastern, central, and western regions, denoted as Region_1 according to the China Health Statistics Yearbook. The other was the eastern, central, western, and northeastern regions, denoted as Region_2 according to the National Bureau of Statistics. Similarly, we also redefined *PeerResTime* and *PeerResInfo* based on the same city and hospital. As shown in columns 1–4 in Table 7, the results are consistent with the main results.

Secondly, physicians in higher-level hospitals often have higher average medical capabilities than those in lower-level hospitals.

Table 6
Robustness Test: model substitution.

Variables	Trend	Count Data Models		Log-Transformation Model
	High-Dimensional Fixed Effect Model	Poisson Model	Negative Binomial Model	Log-model
Free	−0.0190	−0.00552	−0.00396	0.0507***
	(−0.55)	(−0.54)	(−1.62)	(7.20)
Price	−0.153***	−0.0690***	−0.0769***	−0.0640***
	(−7.48)	(−7.93)	(−19.53)	(−10.96)
NOR	0.0729***	0.0156***	0.0107***	0.0191***
	(4.40)	(4.56)	(5.27)	(4.80)
NOG	0.0142	0.00301	0.00175	0.0112***
	(0.62)	(0.73)	(1.25)	(2.77)
NOA	0.0117**	0.00584***	0.00567***	0.00546**
	(2.53)	(2.93)	(2.92)	(2.23)
PeerResTime	−0.0446	0.00696	0.00810	0.00972
	(−1.37)	(0.67)	(1.30)	(1.07)
PeerResInfo	−0.121***	−0.155***	−0.150***	−0.165***
	(−2.19)	(−6.29)	(−10.30)	(−8.33)
FocResTime	−0.0314***	−0.0193***	−0.0187***	−0.0213***
	(−7.94)	(−8.60)	(−7.96)	(−10.61)
FocResInfo	0.0224***	0.00978***	0.0118***	0.0127***
	(3.23)	(3.03)	(4.53)	(4.82)
TFE	YES	YES	YES	YES
IFE	YES	YES	YES	YES
Department-specific linear trends	YES			
_cons	1.858***	0.867***	4.543***	0.148***
	(11.47)	(358.32)	(68.94)	(5.92)
Observations	196,570	196,570	196,570	196,570

Notes: t statistics in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 7
Robustness Test: variable substitution – the range of peer physicians.

Variables	Region1	Region2	City	Hospital	Hospital Level	Title
Free	−0.0118	−0.0119	−0.0121	−0.0121	−0.0116	−0.0116
	(−0.78)	(−0.79)	(−0.80)	(−0.80)	(−0.76)	(−0.76)
Price	−0.0672***	−0.0672***	−0.0671***	−0.0670***	−0.0673***	−0.0673***
	(−7.44)	(−7.44)	(−7.44)	(−7.43)	(−7.45)	(−7.45)
NOA	0.00516**	0.00518**	0.00514**	0.00516**	0.00518**	0.00521**
	(2.52)	(2.53)	(2.51)	(2.52)	(2.53)	(2.54)
NOG	0.00639	0.00639	0.00639	0.00638	0.00640	0.00639
	(0.63)	(0.63)	(0.63)	(0.63)	(0.63)	(0.63)
NOR	0.0325***	0.0326***	0.0326***	0.0326***	0.0325***	0.0326***
	(4.42)	(4.43)	(4.43)	(4.43)	(4.42)	(4.43)
FocResTime	−0.0135***	−0.0135***	−0.0135***	−0.0135***	−0.0134***	−0.0135***
	(−7.97)	(−7.97)	(−7.96)	(−7.98)	(−7.94)	(−7.95)
FocResInfo	0.00910***	0.00912***	0.00910***	0.00903***	0.00919***	0.00932***
	(3.01)	(3.02)	(3.02)	(2.99)	(3.05)	(3.09)
PeerResTime_reg1	0.00356					
	(0.58)					
PeerResInfo_reg1	−0.0397***					
	(−2.76)					
PeerResTime_reg2		0.00210				
		(0.37)				
PeerResInfo_reg2		−0.0333***				
		(−2.96)				
PeerResTime_city			−0.000165			
			(−0.08)			
PeerResInfo_city			−0.0111***			
			(−4.06)			
PeerResTime_hos				0.00307		
				(1.27)		
PeerResInfo_hos				−0.00975***		
				(−4.22)		
PeerResTime_hlevel					−0.00199	
					(−0.20)	
PeerResInfo_hlevel					−0.0592	
					(−2.72)	
PeerResTime_title						−0.00435
						(−0.42)
PeerResInfo_title						−0.0673***
						(−4.19)
TFE	YES	YES	YES	YES	YES	YES
IFE	YES	YES	YES	YES	YES	YES
_cons	0.0142	0.00616	−0.0217**	−0.0243**	0.0285	0.0336
	(0.76)	(0.38)	(−2.06)	(−2.34)	(1.13)	(1.66)
Observations	196,570	196,570	196,570	196,570	196,570	196,570

Notes: t statistics in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 8
Robustness Test: variable substitution – the measurement of informativeness.

Variables	Model 1	Model 2
	VolPay	VolPay
FocResTime	−0.0139***	−0.0140***
	(−7.87)	(−7.91)
FocResInfo_Q&Q	0.00694***	
	(2.64)	
FocReadability		0.00924***
		(3.48)
PeerResTime	−0.00284	−0.00108
	(−0.19)	(−0.07)
PeerResInfo_Q&Q	−0.0755***	
	(−3.21)	
PeerReadability		−0.132***
		(−4.34)
Free	−0.0116	−0.0117
	(−0.75)	(−0.76)
Price	−0.0662***	−0.0682***
	(−7.53)	(−7.62)
NOA	0.00507**	0.00499**
	(2.45)	(2.41)
NOG	0.00628	0.00622
	(0.62)	(0.62)
NOR	0.0331***	0.0330***
	(4.47)	(4.46)
TFE	YES	YES
IFE	YES	YES
_cons	0.0571*	0.110***
	(1.71)	(3.14)
Observations	196,570	196,570

t statistics in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Therefore, we redefined *PeerResTime* and *PeerResInfo* according to the level of the hospital. In this part, hospitals where physicians were located were divided into high level hospitals and non-high level hospitals. As shown in column 5 in Table 7, the results remain consistent with the main results.

Thirdly, the physician's professional title indicates a physician's expertise capability and medical experience. Therefore, we redefined *PeerResTime* and *PeerResInfo* according to physicians' professional titles. We classified physician titles into two categories: high levels and primary level. As shown in column 6 in Table 7, the results are consistent with the main results.

Furthermore, to validate the impact of response informativeness on performance from a quality perspective, referring to Chen et al. [63] and Jin et al. [67], we use quality (Q&Q) and readability as alternative measures of informativeness. The results, as shown in Table 8, support our main findings.

5.4.3. Heterogeneity analysis

While the focal physician's response efforts and colleagues' response informativeness all impact online performance, these effects may vary with medical specialties. Therefore, we investigate how the effects of physicians' response efforts vary across departments (see Figs. 5–8).

As shown in Figs. 5–8, each department's results are consistent with the main findings, except for the Department of Ophthalmology and Otorhinolaryngology, the Department of Stomatology, as well as the departments that address general concerns and other issues. We believe this is not a concern. These departments have a small number of physicians (as illustrated in Fig. 3) and are not mainstream departments. Moreover, our study focuses on the overall average number of physicians on the online platform; thus, the main conclusions remain reliable. Specifically, despite the significant presence of internists, the insignificance of *FocResInfo* may be because internal medicine involves a wide and heterogeneous range of diseases, and there is significant variability among subspecialties. For instance, conditions such as bronchitis, hypertension, diabetes, and peptic ulcers fall under respiratory medicine, cardiology, endocrinology, and gastroenterology, respectively, each representing distinct and unrelated specialties. The impact of *FocResInfo* may vary across these subspecialties and could offset each other to varying degrees. Moreover, patients in internal medicine often describe symptoms extensively, but accurate diagnosis and effective treatment of various conditions typically require instrumental examinations such as ultrasound and X-rays, as well as laboratory tests such as blood tests and genetic screenings. However, in online consultations, physicians rely solely on the textual descriptions provided by patients. This lack of support from instrumental examinations and laboratory tests may result in physicians being unable to diagnose diseases definitively, instead responding such as "please conduct *** tests", independent of the informativeness of the response.

Furthermore, on the platform, physicians either upload personal pictures or disclose their gender in the introduction. Patients can directly find gender information from the profile pictures when they search for a physician, but they must click each physician's page to read the introduction. We divided our sample into two groups based on whether or not the physicians uploaded pictures. The results are shown

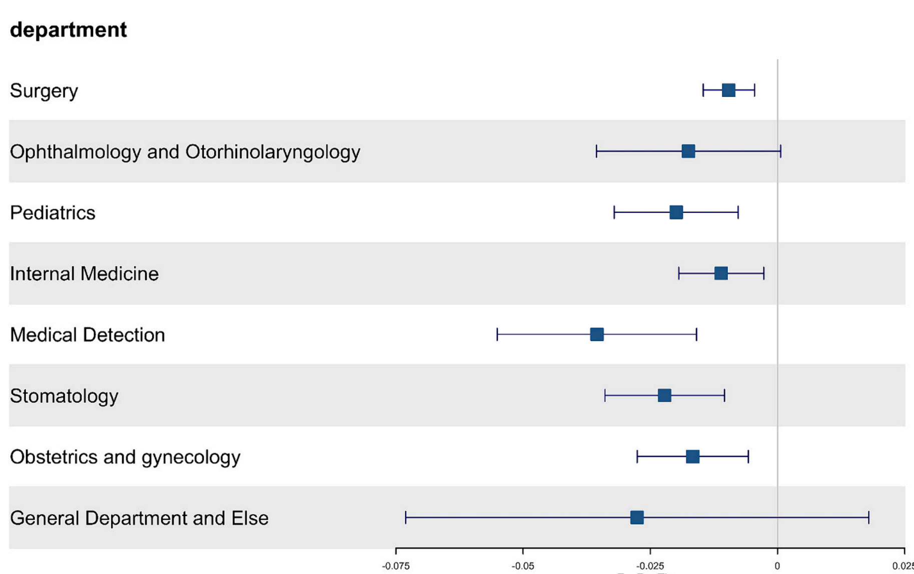

Fig. 5. Heterogeneity Tests of Department – *FocResTime*.

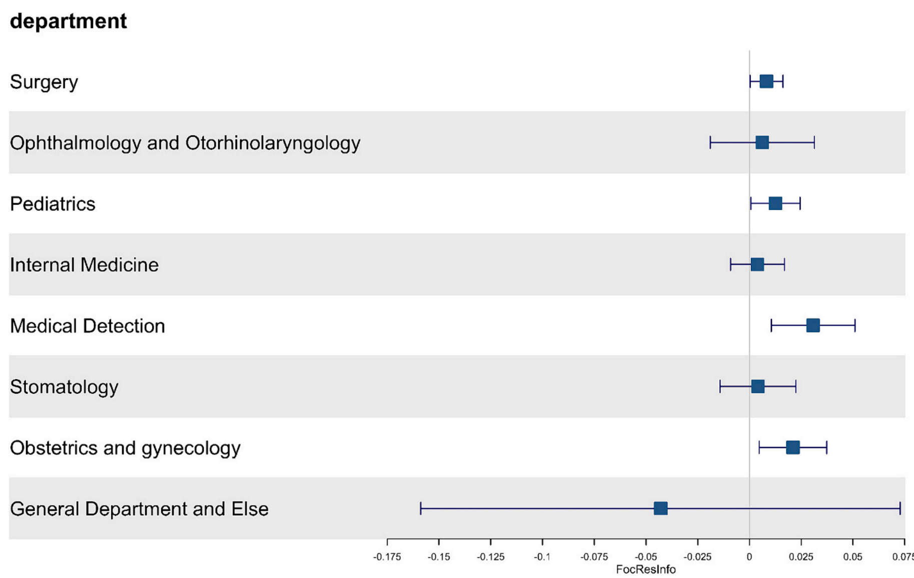

Fig. 6. Heterogeneity Tests of Department - *FocResInfo*.

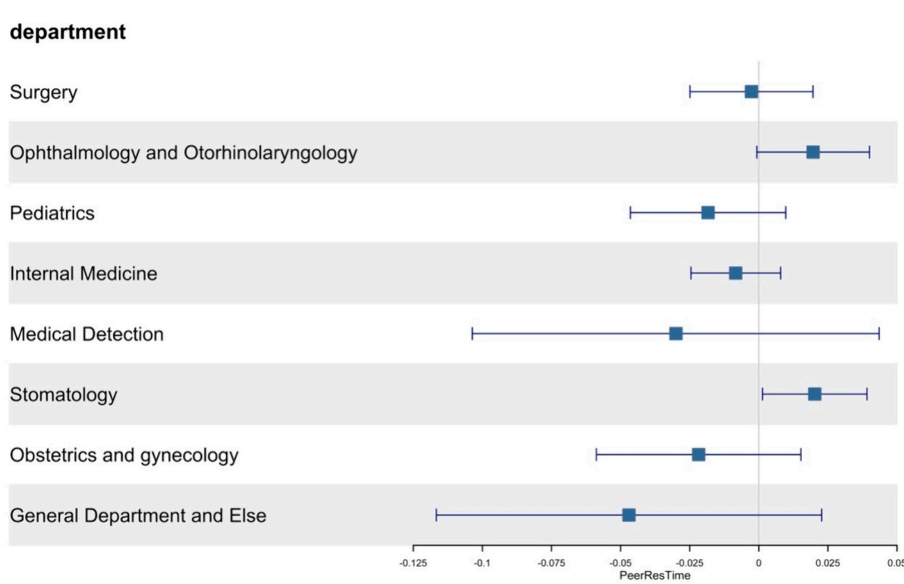

Fig. 7. Heterogeneity Tests of Department – *PeerResTime*.

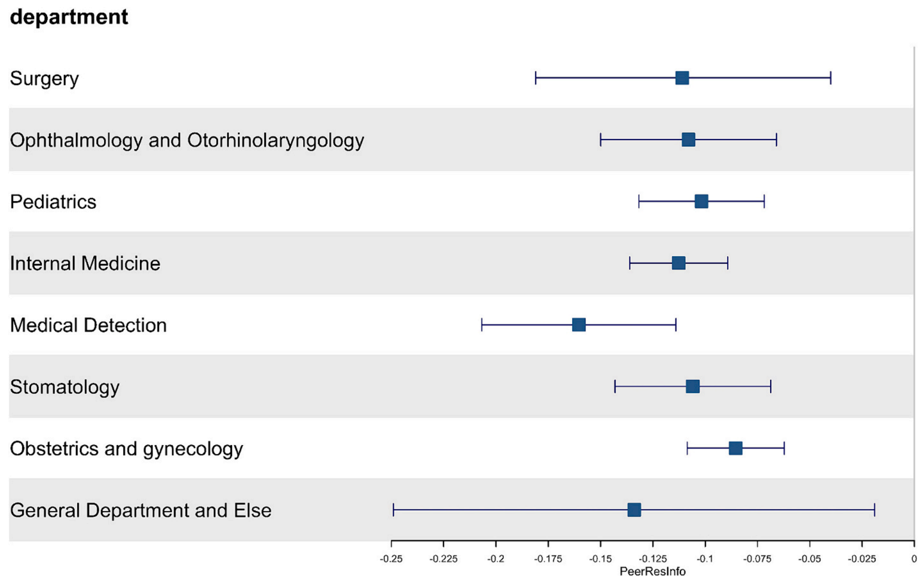

Fig. 8. Heterogeneity Tests of Department - *PeerResInfo*.

Table 9
Heterogeneity Tests for Whether to Upload a Picture (Moderating Effect)

Variables	Model 1	Model 2
	VolPay	VolPay
stdFocResTime	−0.0168***	−0.0143***
	(−5.24)	(−3.06)
stdFocResInfo	0.00548	0.0157***
	(1.15)	(2.61)
stdPeerResTime	−0.0100	0.0244
	(−0.59)	(1.30)
stdPeerResInfo	−0.119***	−0.118**
	(−3.69)	(−2.09)
stdFocResTime × *Male*	0.00644	0.00111
	(1.58)	(0.19)
stdFocResInfo × *Male*	0.00484	−0.00204
	(0.84)	(−0.30)
stdPeerResTime × *Male*	−0.00261	−0.00544
	(−0.26)	(−0.42)
stdPeerResInfo × *Male*	−0.0474***	−0.0272**
	(−4.73)	(−2.11)
Free	−0.0165	0.0274
	(−0.96)	(1.62)
NOA	0.00441**	0.0235**
	(2.20)	(2.31)
NOG	0.00596	0.0199
	(0.56)	(1.36)
NOR	0.0307***	0.0495*
	(4.05)	(1.94)
Price	−0.0631***	−0.102
	(−6.49)	(−4.25)
TFE	YES	YES
IFE	YES	YES
_cons	0.132	0.0560
	(3.58)	(0.69)
Observations	159,418	37,152

t statistics in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

in Table 9. Interestingly, we discovered that the performance of male physicians who uploaded profile pictures was more susceptible to the influence of peer physicians' informativeness than that of male physicians who did not upload profile pictures. The likely reason for this is that physicians who uploaded profile pictures preferred and valued providing healthcare on the platform more, and therefore, they cared more about keeping an eye on how they and their peers performed on the platform and were more inclined to compete.

6. Discussion and implications

Using data collected from the leading OHPs in China, this study extracts service-related information from physicians' response behavior. It investigates whether and how physicians' response effort information affects their online performance from a competitive perspective, as well as gender differences in these effects. First, we find that physician response efforts affect online performance. When the focal physician responds more quickly or with richer information or when peers respond with less information, the focal physician performs better. This finding indicates that response time and informativeness are factors influencing online performance. Second, we find that the informativeness of peer physicians' responses positively affects the focal physician's responses, suggesting a competitive effect among healthcare personnel. Third, our findings further reveal that physician gender plays a role in moderating the relationship between peer physicians' response informativeness and the focal physician's online performance, such that, compared with women, men perform better in providing richer information. We also find heterogeneity among departments. Interestingly, we also discovered that the performance of male physicians who uploaded profile pictures was more susceptible to the influence of peer physicians' informativeness than that of male physicians who did not upload profile pictures.

6.1. Theoretical implications

This study has several theoretical contributions. First, this study contributes to the literature on performance by revealing the effect of physician response efforts on online performance from a competitive perspective. Most previous studies have focused on how physicians' individual behaviors or characteristics affect performance [5,8]. Whether the behavior of colleagues providing similar medical services also affects online performance from a competitive perspective has not been fully considered. Thus, our work not only analyzes at the individual level but also comprehensively considers the potential impact of their competitors. Our research findings indicate that the response efforts of both the focal physician and his or her colleagues affect the focal physician's online performance.

Second, previous studies have examined service competition in offline contexts [19,68], and a more limited exploration of online contexts mainly focused on e-commerce [10,12,14,69]. A key contribution of our research is that we extend competition theory to the online healthcare community literature. Based on competition theory, our study adds to this research by investigating the mediating effects of the focal physician's response efforts between colleagues' response efforts and online performance. Our study provides theoretical evidence that colleagues' response efforts affect online performance by influencing the focal physician's response efforts and that the degree of a physician's online service efforts positively impacts patient decision-making. Moreover, compared to previous research, this paper uses more detailed variables, response time and response informativeness, to explore the mechanism through which the focal physician's and competitors' response efforts affect online performance.

Third, our work advances the research on gender differences in online services by examining how gender moderates the association between physician response efforts and online performance. Previous research has established the significance of gender effects in online services. We take a step further by examining response time and response informativeness as distinct types of service behaviors. Despite the well-documented importance of gender differences, research conclusions remain inconsistent [27,28], and little is known about how gender affects the impact of different types of behavior on online performance. Consistent with prior studies [29,30], our study proves that male physicians prefer competition. However, in the healthcare context, while male physicians may prefer competition, their competitive advantage is less than that of female physicians.

6.2. Practical implications

Our study also has implications for healthcare providers and platforms. First, we observe that physicians' response efforts significantly influence their performance and that there is competition among physicians on OHPs. To achieve better performance, physicians should make decisions that enhance response speed or increase the informativeness of their responses. This result provides physicians with decision support on how to contribute to improving their service quality. Interestingly, however, changes in colleagues' response time do not significantly affect online performance. A decrease in performance is noticeable only when colleagues respond with richer information. Physicians should monitor the response informativeness of their peers on the platform to compete effectively and make their own response decisions based on their colleagues' responses. A physician's online performance may not reach its optimum without considering peers' response informativeness. These findings provide information about the existence and impact of competitors, helping physicians identify avenues to enhance their online reputation and core competitiveness and make response decisions, ultimately leading to better performance.

Although patients, physicians, and platform administrators are aware of the above rules, it is challenging to quantify physicians' response efforts intuitively, visually, rapidly, and accurately in practice. Therefore, we recommend that platforms add a "response efforts index" on each physician's homepage that includes individual response time and response informativeness within the current week, as well as the average response time and informativeness of peer physicians, to provide an important reference for evaluating online physician services. Adding this index would enrich the pre-consultation information available to patients, helping them have a more comprehensive and detailed understanding of physicians and guiding them to make better decisions about the adoption of physicians [3,63]. At the same time, the index provides timely and comprehensive feedback on the response efforts of physicians and their competitors, helping physicians adjust and improve their response behavior and adjust their decision-making in line with their actual situation. Our findings also provide important insights for platforms to assess and effectively manage online physician services. Although our findings are derived from the Chinese medical system, they can also be applied to similar medical settings.

Second, our findings will help physicians formulate appropriate online response strategies considering gender-based influences on physicians' performance. The results show that peer physicians' response informativeness has a stronger impact on the performance of male focal physicians than on that of female focal physicians. When the informativeness of peer responses decreases, the online performance of male physicians decreases more significantly than that of female physicians. That is, while male physicians may prefer competition, their competitive advantage is less than that of female physicians. Male physicians must be reminded to remain vigilant and maintain their response informativeness when their peers' response informativeness decreases. Conversely, female physicians should be more proactive when peers' informativeness increases. More interestingly, the performance of male physicians who uploaded profile pictures is more susceptible to the influence of peer physicians' informativeness than that of male physicians who did not upload profile pictures. This finding suggests that physicians should make the decision to upload their personal pictures if they prefer to compete. Finally, our research results can be extended to similar service industries, such as open-source communities.

6.3. Limitations and future research

There are also limitations of this study. First, the analysis was performed at the physician level, controlling for the influence of the individual heterogeneity of physicians. In the future, patient-level data can be included in the model. Second, this research considered only online factors. However, the impact of physicians' offline word-of-mouth and their ability to acquire customers offline has not yet been considered. Most patients have offline visits with online follow-ups, postdiagnosis check-ins, or consultations, so their consultation and selection behaviors are not purely online. Therefore, future research may consider combining physician online efforts and reputation with offline efforts and word-of-mouth to examine their joint impact on physicians' customer acquisition ability, as well as the interplay between online and offline factors. Finally, due to the lack of data, our model does not account for new physicians who may have few paid consults to start just as a result of their newness. Future research may explore how rapidly they establish themselves to be more comparable to the majority of physicians in the study.

CRediT authorship contribution statement

Haoyu Ren: Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Data curation, Conceptualization. **Liuan Wang:** Writing – review & editing, Validation, Methodology, Data curation, Conceptualization. **Junjie Wu:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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